

# Echo-Corrected Information Diffusion: Certified Multiscale Phase-Type Hawkes Processes for Macro-News Absorption across FX, Equities, Gold and Crypto

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## Abstract

How fast do prices absorb scheduled macroeconomic news, and how much of the post-release activity is the market reacting to itself? We study six markets (EUR/USD, USD/JPY, gold, the S&P 500, bitcoin and ether) around 670 scheduled releases between 2022 and 2026. Price changes are recorded as  $\delta$ -crossing events and modelled with a multiscale phase-type Hawkes process whose parameters all enter linearly. The log-likelihood is then concave, and we compute the global maximum with a Fenchel duality-gap certificate. A Markov embedding gives the echo-corrected response to a release in closed form; we prove it is stable iff the spectral radius of the branching matrix is below one. A built-in placebo test on matched release-free days, with a penalty chosen on held-out windows, separates genuine news effects from artefacts: CPI, payrolls, PPI, the FOMC statement and retail sales pass it, second-tier releases and the ECB and BoJ do not. For the releases that pass, the direct reaction is absorbed in about a second, but echoes stretch half of the response to 16–137s and ninety percent to 5–11 minutes; the branching ratio is  $\rho(G) = 0.87$  and declines from 2022 to 2026. On synthetic benchmarks the certified estimator recovers multiscale and network kernels with 7–10 $\times$  smaller likelihood deficits than six classical estimators. We also introduce EPT-X, a neural point process with an exact compensator that superposes a history-conditioned hyper-Erlang renewal channel on a linear Hawkes backbone; on LOBSTER limit-order-book data it achieves the best test likelihood on all five stocks against six classical estimators and seven neural baselines, by 0.23–0.45 nats per event over the best neural model. On the public EasyTPP benchmarks it improves the best published likelihood on Retweet by more than 0.6 nats per event and matches it on Taobao and StackOverflow. Along the way we find that a widely used IntensityFree implementation clamps inter-event times before scoring, which overstates its likelihood on streams with many short gaps.

## 1 Introduction

Price changes, order-book messages and trades arrive as streams of events in continuous time, and temporal point processes are the standard way to model them. They are used for two rather different purposes. In econometrics, Hawkes processes [10] are used to *measure* market mechanisms. Their branching structure separates exogenous events, such as a reaction to news, from endogenous events triggered by earlier trading, and the fitted kernels describe how quickly information is absorbed [8, 9, 14]. In machine learning, neural temporal point processes are used to *predict* the next event, and they are compared by their held-out likelihood on shared benchmarks [5, 7, 12, 15, 22, 25].

Each use has a weakness. Measurement relies on non-convex fits whose optimality is unknown, and on summaries that ignore part of the response. A common example is the “half-life”  $\ln 2/\beta$  of an exponential news kernel with decay rate  $\beta$ , which describes the direct reaction to news but not the further trading that this reaction sets off. On the prediction side, most intensity-based neural models estimate the compensator  $\int \lambda(t) dt$  by Monte Carlo [5, 12, 23, 25], so their reported likelihoods contain sampling error. Models with an exact likelihood do exist. FullyNN [13] models the cumulative hazard with a monotone network, and IntensityFree [15] models the density of the next gap directly. Both, however, give up the Hawkes structure that makes a model interpretable [16]. We also find that reported likelihoods can depend on implementation details.

In this paper we show that a single algebraic object serves both purposes with exact likelihoods. We call it a *phase-type state space*: every kernel is a nonnegative combination of Erlang densities whose rates lie on a logarithmic grid of time scales. Between two events such a state evolves by an explicit matrix exponential, so intensities and their integrals are available in closed form. The class is also rich, since phase-type densities are dense in the set of densities on  $\mathbb{R}_+$ .

## Contributions.

1. **Certified measurement (MSX).** We define a multiscale phase-type Hawkes process with marked news kernels in which every parameter enters the intensity linearly. Its log-likelihood is therefore concave. We compute the global maximum with accelerated EM followed by a barrier Newton step, and each fit comes with a Fenchel duality gap that bounds its distance from the optimum (Proposition 2).
2. **Echo-corrected responses in closed form.** Embedding the process in a linear system gives the mean response to a shock as  $Ce^{At}z_0$ . We prove that this system is stable exactly when the spectral radius of the branching matrix is below one (Theorem 3), and we obtain absorption times, echo amplification and price-discovery curves without simulation.
3. **Exact neural prediction (EPT-X).** We introduce a neural point process whose latent state is phase-type. It combines a linear Hawkes backbone with a history-dependent renewal channel, a mixture of Erlang distributions that also allows “no event”, added as a competing risk. The likelihood remains exact, the model contains the Hawkes backbone as a special case, and the renewal channel can approximate any conditional distribution of the next gap (Proposition 5).
4. **Benchmarks and a correction to one baseline.** EPT-X has the best likelihood on all five LOBSTER order-book streams and on Retweet, and it is within one standard deviation of the best published model on Taobao and StackOverflow. We also find that a widely used IntensityFree implementation replaces every inter-event time below  $10^{-5}$  s by  $10^{-5}$  s before scoring. Scored on the observed gaps, it loses its lead on two order-book streams.
5. **Application.** We study six markets around 670 scheduled macroeconomic releases between 2022 and 2026, with a placebo test built into the likelihood. The direct reaction to CPI, payrolls, PPI and the FOMC statement is absorbed in about a second, but the market’s subsequent trading stretches half of the response to 16–137 s and ninety percent of it to 5–11 minutes.

The ingredients are not new on their own. Linear-in-parameters kernel dictionaries [2], EM algorithms for Hawkes processes [21, 24], Erlang-chain embeddings and exact-likelihood neural

point processes [13, 15] have all been studied. What we add is their combination: certified, closed-form measurement on one side, an exact neural model that contains the linear Hawkes process on the other, and experiments showing that both work.

## 2 The MSX-Hawkes model

**Events.** For each asset  $a$  we record  $\delta$ -crossing events. An upward (downward) event occurs whenever the log mid-price has risen (fallen) by  $\delta_a$  since the previous event. If a single price change crosses several levels, we record one event at that time and keep the number of levels crossed as a size mark. This sampling in intrinsic time filters out bid-ask bounce and makes assets comparable in basis points. It also links intensities to prices: over any interval, the change in log price equals  $\delta_a$  times the number of upward minus downward levels crossed. Event type  $2a$  counts upward crossings of asset  $a$  and type  $2a + 1$  downward ones, so  $A$  assets give  $d = 2A$  event types.

**A dictionary of Erlang densities.** Fix a set of rates  $\beta_1, \dots, \beta_K > 0$  on a logarithmic grid, so that each rate corresponds to a time scale  $1/\beta_k$ , and an Erlang order  $R \geq 1$ . For  $k = 1, \dots, K$  and  $r = 0, \dots, R - 1$  we define the basis densities

$$g_{kr}(t) = \beta_k \frac{(\beta_k t)^r}{r!} e^{-\beta_k t}, \quad t \geq 0. \quad (1)$$

Each  $g_{kr}$  is the density of a gamma (Erlang) distribution with shape  $r + 1$  and rate  $\beta_k$ , so  $\int_0^\infty g_{kr}(t) dt = 1$  and its mean is  $(r + 1)/\beta_k$ . A kernel is a nonnegative combination

$$\phi(t) = \sum_{k=1}^K \sum_{r=0}^{R-1} a_{kr} g_{kr}(t), \quad a_{kr} \geq 0. \quad (2)$$

Because each basis density integrates to one, the total mass of the kernel, which is its branching ratio, is simply  $\sum_{k,r} a_{kr}$ . With  $R = 1$  the dictionary contains only exponentials, and mixtures of exponentials can approximate any completely monotone kernel, power laws included (Bernstein's theorem). Orders  $r \geq 1$  add kernels that start at zero and peak later, which describe delayed reactions. More generally, phase-type densities are dense in the set of densities on  $\mathbb{R}_+$ .

**Model.** We observe  $d$  event types (for example upward and downward price moves of several assets) inside observation windows  $w$ , each with a likelihood interval  $[t_0^w, t_1^w)$ . Exogenous shocks  $e$ , here scheduled macroeconomic releases, occur at known times  $\tau_e$ . Each shock has a type  $c_e$  (CPI, payrolls, and so on) and a nonnegative mark vector  $\mathbf{z}_e = (1, z_e^+, z_e^-)$ , where  $z_e^+$  and  $z_e^-$  are the positive and negative parts of the standardised surprise. The MSX intensity of event type  $i$  is

$$\begin{aligned} \lambda_i(t) = & \underbrace{c_{iw} + \sum_m \eta_{im} h_m(\text{tod}(t))}_{\text{baseline}} + \underbrace{\sum_{j=1}^d \sum_{n: t_n^j < t} \sum_{k,r} a_{ijk r} g_{kr}(t - t_n^j)}_{\text{endogenous}} \\ & + \underbrace{\sum_{e: \tau_e < t} \sum_{k,r,m} b_{ic_e k r m} z_{em} g_{kr}^{\text{exo}}(t - \tau_e)}_{\text{response to news}} + \underbrace{\sum_{e: \tau_e > t} \sum_k \alpha_{ic_e k} \beta_k^{\text{ant}} e^{-\beta_k^{\text{ant}}(\tau_e - t)}}_{\text{anticipation}}. \end{aligned} \quad (3)$$

The terms are as follows.

- *Baseline.*  $c_{iw}$  is an intercept for window  $w$ , and  $h_m$  are periodic hat functions of the local time of day  $\text{tod}(t)$  with weights  $\eta_{im}$ .
- *Endogenous excitation.*  $t_n^j$  is the  $n$ -th event of type  $j$ , the  $g_{kr}$  are the basis densities (1), and  $a_{ijk_r}$  is the weight with which an event of type  $j$  excites type  $i$  at time scale  $k$  and order  $r$ .
- *Response to news.*  $g_{kr}^{\text{exo}}$  is a second dictionary with its own rates, and  $b_{ickrm}$  weights mark component  $m$  of a release of type  $c$ .
- *Anticipation.*  $\alpha_{ick}$  captures activity that builds up before a release of type  $c$ , with rates  $\beta_k^{\text{ant}}$ .

All parameters are nonnegative. Every term is linear in its parameters, so we can write  $\lambda_i(t) = \mathbf{x}(t)^\top \boldsymbol{\theta}_i$ , where  $\boldsymbol{\theta}_i$  stacks the parameters of type  $i$  and  $\mathbf{x}(t) \geq 0$  stacks the corresponding known functions of time. The log-likelihood then separates across event types:

$$\ell_i(\boldsymbol{\theta}_i) = \sum_{n: u_n=i} \log(\mathbf{x}(t_n)^\top \boldsymbol{\theta}_i) - \boldsymbol{\Phi}^\top \boldsymbol{\theta}_i, \quad \boldsymbol{\Phi} = \sum_w \int_{t_0^w}^{t_1^w} \mathbf{x}(t) dt, \quad (4)$$

where  $u_n$  is the type of the  $n$ -th event. Every entry of  $\mathbf{x}(t_n)$  and of  $\boldsymbol{\Phi}$  follows from Lemma 1.

## 3 Theory

### 3.1 Exact recursion

**Exact recursion.** Every state in this paper is built from such Erlang chains: the Hawkes excitation, the response to news, the neural latent state of EPT and the renewal channel of EPT-X. For all of them, the evolution between two events and its integral are available in closed form. The following lemma gives the formulas for one source and one rate.

**Lemma 1** (Erlang-chain recursion). *Let  $t_1 < t_2 < \dots$  be the event times of one source, let  $\beta > 0$  be one rate of the dictionary, and define the chain of states*

$$s_r(t) = \sum_{n: t_n < t} g_r(t - t_n), \quad r = 0, \dots, R-1,$$

where  $g_r$  is the basis density (1) with rate  $\beta$ . Write  $\mathbf{s} = (s_0, \dots, s_{R-1})^\top$ , let  $I$  be the  $R \times R$  identity and let  $B$  be the  $R \times R$  down-shift matrix ( $B_{r,r-1} = 1$  and all other entries zero).

1. Between events,  $\dot{\mathbf{s}} = \beta(B - I) \mathbf{s}$ ; at an event,  $s_0$  increases by  $\beta$ .
2. For a gap  $\Delta \geq 0$  with no event in  $(t, t + \Delta)$ , let  $y = \beta\Delta$  and  $w_j = e^{-y} y^j / j!$ . Then

$$s_r(t + \Delta) = \sum_{j=0}^r w_j s_{r-j}(t), \quad \int_t^{t+\Delta} s_r(u) du = \frac{1}{\beta} \sum_{j=0}^r P(j+1, y) s_{r-j}(t),$$

where  $P(a, y)$  is the regularised lower incomplete gamma function.

*Proof. Step 1 (dynamics).* Differentiating (1) gives

$$\frac{d}{dt} g_0(t) = -\beta g_0(t), \quad \frac{d}{dt} g_r(t) = \beta(g_{r-1}(t) - g_r(t)) \quad (r \geq 1).$$

Summing over past events gives  $\dot{s}_0 = -\beta s_0$  and  $\dot{s}_r = \beta(s_{r-1} - s_r)$ , which is  $\dot{\mathbf{s}} = \beta(B - I)\mathbf{s}$ . Since  $g_0(0) = \beta$  and  $g_r(0) = 0$  for  $r \geq 1$ , a new event adds  $\beta$  to  $s_0$  and leaves the other states unchanged.

*Step 2 (evolution over a gap).* The solution is  $\mathbf{s}(t + \Delta) = e^{\beta(B-I)\Delta}\mathbf{s}(t)$ . The matrices  $B$  and  $I$  commute and  $B^R = 0$ , so

$$e^{\beta(B-I)\Delta} = e^{-y} e^{yB} = e^{-y} \sum_{j=0}^{R-1} \frac{y^j}{j!} B^j.$$

Because  $B^j$  shifts a vector down by  $j$  places, component  $r$  of  $e^{\beta(B-I)\Delta}\mathbf{s}(t)$  is  $\sum_{j \leq r} w_j s_{r-j}(t)$ .

*Step 3 (integral over a gap).* Integrating the weights over the gap,

$$\int_0^\Delta e^{-\beta u} \frac{(\beta u)^j}{j!} du = \frac{1}{\beta} P(j+1, y),$$

which gives the second formula.  $\square$

Evaluating a likelihood therefore costs  $O(NdKR^2)$  operations for  $N$  events in  $d$  dimensions, and it involves no Monte Carlo or quadrature error.

### 3.2 Concavity, EM and a certificate of global optimality

**Concavity and a certificate.** Problem (4) maximises a concave function over the nonnegative orthant. We add an  $\ell_1$  penalty  $\gamma^\top \boldsymbol{\theta}_i$  with weights  $\gamma \geq 0$  proportional to each feature's exposure; since  $\boldsymbol{\theta}_i \geq 0$ , the penalty is linear and simply adds  $\gamma$  to  $\boldsymbol{\Phi}$ . We maximise with the multiplicative EM update

$$\theta_p \leftarrow \theta_p \frac{\sum_n x_{np}/\lambda_n}{\Phi_p + \gamma_p},$$

where  $x_{np}$  is feature  $p$  at the  $n$ -th event of type  $i$  and  $\lambda_n$  the current intensity there. We accelerate it with SQUAREM [18] and finish with a log-barrier Newton method. The following bound tells us how far any fit is from the optimum.

**Proposition 2** (Duality-gap certificate). *Fix an event type and drop the index  $i$ . Let  $X$  be the matrix whose  $n$ -th row is  $\mathbf{x}(t_n)^\top$  for the  $N$  events of this type, let  $\boldsymbol{\theta} \geq 0$  be any parameter vector with intensities  $\lambda_n = (X\boldsymbol{\theta})_n > 0$ , and let  $\ell^*$  be the maximum of the penalised log-likelihood  $\ell(\boldsymbol{\theta}) = \sum_n \log \lambda_n - (\boldsymbol{\Phi} + \boldsymbol{\gamma})^\top \boldsymbol{\theta}$ . Define*

$$s = \max_p \frac{\sum_n x_{np}/\lambda_n}{\Phi_p + \gamma_p}.$$

*Then*

$$\ell^* - \ell(\boldsymbol{\theta}) \leq N \log s + (\boldsymbol{\theta}^\top (\boldsymbol{\Phi} + \boldsymbol{\gamma}) - N),$$

*and both terms on the right are zero at the optimum.*

*Proof.* We use the notation of the proposition:  $X$  has rows  $\mathbf{x}_n^\top = \mathbf{x}(t_n)^\top$ ,  $\lambda_n = \mathbf{x}_n^\top \boldsymbol{\theta}$ , and  $\ell(\boldsymbol{\theta}) = \sum_{n=1}^N \log \lambda_n - (\boldsymbol{\Phi} + \boldsymbol{\gamma})^\top \boldsymbol{\theta}$ .

*Step 1 (an upper bound for every feasible point).* For  $u, v > 0$ , Fenchel's inequality gives

$$\log u \leq v u - 1 - \log v.$$

Take any vector  $\mathbf{v} = (v_1, \dots, v_N)$  with positive entries such that  $X^\top \mathbf{v} \leq \Phi + \gamma$  componentwise. Applying the inequality with  $u = \lambda_n$  and  $v = v_n$  and summing over  $n$ ,

$$\sum_n \log \lambda_n \leq \sum_n v_n \lambda_n - N - \sum_n \log v_n = \boldsymbol{\theta}^\top X^\top \mathbf{v} - N - \sum_n \log v_n.$$

Since  $\boldsymbol{\theta} \geq 0$  and  $X^\top \mathbf{v} \leq \Phi + \gamma$ , we have  $\boldsymbol{\theta}^\top X^\top \mathbf{v} \leq \boldsymbol{\theta}^\top (\Phi + \gamma)$ . Subtracting  $(\Phi + \gamma)^\top \boldsymbol{\theta}$  from both sides therefore gives

$$\ell(\boldsymbol{\theta}) \leq D(\mathbf{v}) := -N - \sum_n \log v_n.$$

This holds for every feasible  $\boldsymbol{\theta}$ , in particular for the maximiser, so  $\ell^* \leq D(\mathbf{v})$ .

*Step 2 (a computable choice of  $\mathbf{v}$ ).* Let  $\boldsymbol{\theta}$  be the current fit and set  $v_n = 1/(s\lambda_n)$ , with  $s = \max_p (\sum_n x_{np}/\lambda_n)/(\Phi_p + \gamma_p)$ . Then  $(X^\top \mathbf{v})_p = \frac{1}{s} \sum_n x_{np}/\lambda_n \leq \Phi_p + \gamma_p$  by the definition of  $s$ , so  $\mathbf{v}$  is admissible in Step 1. For this choice,

$$D(\mathbf{v}) = -N + N \log s + \sum_n \log \lambda_n, \quad \text{hence} \quad \ell^* - \ell(\boldsymbol{\theta}) \leq D(\mathbf{v}) - \ell(\boldsymbol{\theta}) = N \log s + \boldsymbol{\theta}^\top (\Phi + \gamma) - N.$$

*Step 3 (the bound vanishes at the optimum).* The Karush–Kuhn–Tucker conditions of the maximisation state that  $\sum_n x_{np}/\lambda_n = \Phi_p + \gamma_p$  when  $\theta_p > 0$  and  $\sum_n x_{np}/\lambda_n \leq \Phi_p + \gamma_p$  when  $\theta_p = 0$ . Hence  $s = 1$ , so  $N \log s = 0$ . Moreover,

$$\boldsymbol{\theta}^\top (\Phi + \gamma) = \sum_p \theta_p \sum_n \frac{x_{np}}{\lambda_n} = \sum_n \frac{\mathbf{x}_n^\top \boldsymbol{\theta}}{\lambda_n} = N,$$

so the second term is also zero. □

Every fit reported in this paper comes with this bound.

### 3.3 Markov embedding and echo-corrected responses

**Markov embedding.** To study the response to a single release, we collect all endogenous Erlang states  $s_{jkr}$  into one vector  $\mathbf{u}(t)$  and the states of the release's news kernels into  $\mathbf{v}(t)$ . Let  $F$  be the block-diagonal generator of the endogenous chains (one block  $\beta_k(B - I)$  per source and rate, as in Lemma 1) and  $F_e$  the generator of the news chains. Let  $G_{\text{in}}$  be the matrix that adds  $\beta_k$  to the first phase of every chain of source  $j$  when an event of type  $j$  occurs. Finally, let  $\Theta$  hold the endogenous weights  $a_{ijk_r}$  and  $\Theta_e$  the news weights of this release, so that the excess intensity is  $\Theta \mathbf{u}(t) + \Theta_e \mathbf{v}(t)$ .

Taking expectations, each event of type  $j$  occurs at the mean rate  $m_j(t)$  and adds  $G_{\text{in}} \mathbf{e}_j$  to  $\mathbf{u}$ . The mean excess intensity  $\mathbf{m}(t)$  caused by the release therefore satisfies a linear system:

$$\dot{\mathbf{z}} = \mathcal{A} \mathbf{z}, \quad \mathcal{A} = \begin{pmatrix} F + G_{\text{in}} \Theta & G_{\text{in}} \Theta_e \\ 0 & F_e \end{pmatrix}, \quad \mathbf{m}(t) = C \mathbf{z}(t), \quad C = \begin{pmatrix} \Theta & \Theta_e \end{pmatrix}, \quad (5)$$

where  $\mathbf{z} = (\bar{\mathbf{u}}, \mathbf{v})$  stacks the expected endogenous states and the news states, and  $\mathbf{z}(0) = \mathbf{z}_0 = (\mathbf{0}, \mathbf{v}_0)$  with  $\mathbf{v}_0$  the news states just after the release.

**Theorem 3 (Stability).** *Let  $G$  be the  $d \times d$  branching matrix with entries  $G_{ij} = \sum_{k,r} a_{ijk_r}$ , the expected number of type- $i$  events triggered directly by one type- $j$  event, and let  $\rho(G)$  be its spectral radius. Then the matrix  $F + G_{\text{in}} \Theta$  is Hurwitz (all eigenvalues have negative real part) if and only if  $\rho(G) < 1$ .*

*Proof. Step 1.* Every block  $\beta_k(B - I)$  of  $F$  is lower triangular with diagonal  $-\beta_k < 0$  and nonnegative off-diagonal entries. Hence  $F$  is Hurwitz with nonnegative off-diagonal entries, and  $-F$  is a nonsingular M-matrix. The matrix  $P = G_{\text{in}}\Theta$  is entrywise nonnegative.

*Step 2.* For such a pair,  $F + P$  is Hurwitz if and only if  $\rho((-F)^{-1}P) < 1$ . This is the regular-splitting theorem for M-matrices [19, Thm. 3.29], applied to  $-(F + P) = (-F) - P$ .

*Step 3.* The products  $AB$  and  $BA$  have the same nonzero eigenvalues, so

$$\rho((-F)^{-1}G_{\text{in}}\Theta) = \rho(\Theta(-F)^{-1}G_{\text{in}}).$$

*Step 4.* Column  $j$  of  $(-F)^{-1}G_{\text{in}} = \int_0^\infty e^{Ft}G_{\text{in}}dt$  contains, for every chain of source  $j$ , the integral over time of the state left by a single type- $j$  event. By Lemma 1 this state is  $g_{kr}(t)$ , whose integral is one. Hence

$$(\Theta(-F)^{-1}G_{\text{in}})_{ij} = \sum_{k,r} a_{ijk} \int_0^\infty g_{kr}(t) dt = \sum_{k,r} a_{ijk} = G_{ij}.$$

Combining Steps 2 to 4,  $F + G_{\text{in}}\Theta$  is Hurwitz if and only if  $\rho(G) < 1$ .  $\square$

**Corollary 4** (Closed-form absorption). *Assume  $\rho(G) < 1$ . Then  $\mathcal{A}$  is Hurwitz, because it is block triangular and  $F_e$  is Hurwitz, and the following quantities are available in closed form.*

1. *The mean excess intensity after a release, echoes included:  $\mathbf{m}(t) = Ce^{At}\mathbf{z}_0$ .*
2. *Its total, the expected number of extra events:  $M = \int_0^\infty \mathbf{m}(t) dt = -CA^{-1}\mathbf{z}_0$ .*
3. *The fraction of the response absorbed by time  $t$ , summed over event types:*

$$\Pi(t) = \frac{\mathbf{1}^\top CA^{-1}(e^{At} - I)\mathbf{z}_0}{-\mathbf{1}^\top CA^{-1}\mathbf{z}_0}.$$

*Its quantiles  $t_q = \Pi^{-1}(q)$  are the echo-corrected absorption times; we report  $t_{50}$  and  $t_{90}$ .*

4. *The amplification  $\kappa = M/M_{\text{direct}}$ , where  $M_{\text{direct}} = -\Theta_e F_e^{-1}\mathbf{v}_0$  is the total of the direct news response  $\mathbf{m}_{\text{direct}}(t) = \Theta_e e^{F_e t}\mathbf{v}_0$  alone.*
5. *For an asset whose price is observed through upward and downward crossings of size  $\delta$  (types  $u$  and  $v$ ), the expected price change up to time  $t$ :  $\delta \int_0^t (m_u(s) - m_v(s)) ds$ .*

The usual “news half-life”  $\ln 2/\beta$  describes only  $\mathbf{m}_{\text{direct}}$ . When  $\rho(G)$  is close to one, the echoes dominate the response, and Section 7 shows that the two can differ by one to two orders of magnitude.

### 3.4 Closed-form conditional forecasts

The same system gives forecasts. Suppose a release occurs at time  $\tau$ , and let  $\mathcal{F}_\tau$  denote the information available just after it: the history of events, the type of the release and its surprise. We compute the endogenous states  $\mathbf{u}(\tau)$  exactly from the history with Lemma 1, set the news states  $\mathbf{v}_0$  from the surprise, and write  $\mathbf{z}_0 = (\mathbf{u}(\tau), \mathbf{v}_0)$ . Over a short horizon we treat the baseline intensity as a constant vector  $\boldsymbol{\mu}$ . Baseline events then also excite the endogenous chains, which adds the constant input  $\mathbf{b} = (G_{\text{in}}\boldsymbol{\mu}, \mathbf{0})$  to the system, so that  $\dot{\mathbf{z}} = \mathcal{A}\mathbf{z} + \mathbf{b}$ . Solving this equation and integrating over  $(\tau, \tau + h]$  gives the expected number of events of each type,

$$\mathbb{E}[N(\tau, \tau + h) \mid \mathcal{F}_\tau] = \boldsymbol{\mu}h + C \left[ \mathcal{A}^{-1}(e^{\mathcal{A}h} - I)\mathbf{z}_0 + \mathcal{A}^{-1}(\mathcal{A}^{-1}(e^{\mathcal{A}h} - I) - hI)\mathbf{b} \right].$$

Because each event corresponds to a price move of size  $\delta_a$ , this is also a forecast of realised variance ( $\delta_a^2$  times the number of events) and, through the difference between upward and downward events, of the price drift (Track D).

## 4 EPT and EPT-X: neural point processes with an exact compensator

We now turn to prediction. Events arrive at times  $t_1 < t_2 < \dots$  with marks  $y_n \in \{1, \dots, M\}$ , and we write  $\Delta_n = t_n - t_{n-1}$  for the gaps. For a time  $t$  after the  $n$ -th event,  $s = t - t_n$  denotes the time elapsed since that event.

**EPT: a neural phase-type state.** The latent state  $\boldsymbol{\xi}(t) \in \mathbb{R}_{\geq 0}^P$  consists of  $L$  channels of Erlang chains of order  $R$ , each with its own learnable rate. Between events it evolves exactly as in Lemma 1. At each event, a recurrent network summarises the history: a GRU [6] receives an embedding of the mark  $y_n$  and log transforms of the gap  $\Delta_n$  and produces a hidden vector  $\mathbf{h}_n$ . In EPT-X the gap enters through its standardised logarithm, which keeps nanosecond-scale gaps distinguishable from zero. The state then jumps to

$$\boldsymbol{\xi}(t_n^+) = \sigma(W_g \mathbf{h}_n) \odot \boldsymbol{\xi}(t_n^-) + \text{softplus}(W_\xi \mathbf{h}_n),$$

where  $\sigma$  is the logistic function,  $\odot$  the elementwise product and  $W_g, W_\xi$  learned matrices. The first term lets the network forget part of the old state; the second adds a nonnegative jump. For  $t \in (t_n, t_{n+1}]$  the intensity of mark  $y$  is

$$\lambda_y^{\text{EPT}}(t) = \mu_y + \sum_{p=1}^P C_{yp} \xi_p(t) + g_{1y} e^{-\alpha s} + g_{2y} (1 - e^{-\kappa s}) + \sum_q \Theta_{yq} \xi_q^{\text{H}}(t). \quad (6)$$

The quantities  $\mu_y, C_{yp}, g_{1y}, g_{2y}, \alpha$  and  $\kappa$  are nonnegative outputs of the network computed from  $\mathbf{h}_n$ , so they stay fixed until the next event. The two hazard terms let the intensity decay or rise with  $s$ , but stay bounded. An unbounded Gompertz term would overflow on heavy-tailed gaps. The last term is a *linear multivariate phase-type Hawkes process*:  $\boldsymbol{\xi}^{\text{H}}(t)$  collects Erlang chains on a fixed logarithmic grid of rates, one group per source mark, and every event of mark  $y$  adds  $\beta_k$  to the first phase of the chains of source  $y$ . Its weights  $\Theta_{yq} \geq 0$  are learned. This term is a residual path: when the neural read-outs are zero, EPT is the Hawkes process of Section 2, so the network only has to learn departures from linear excitation.

### 4.1 EPT-X: a renewal channel

Order-book and social streams contain cascades, several events in a very short interval followed by a pause. A linear Hawkes intensity describes excitation *between* cascades well, but it has no direct handle on the distribution of gaps *within* them. EPT-X therefore adds a second channel that models the next gap directly.

We fix  $Q$  Erlang “atoms” indexed by  $q$ , each with an order  $R_q$  and a rate  $\beta_q$ . Their means  $R_q/\beta_q$  lie on a logarithmic grid that spans the time scales of the data, and their orders come from  $\{1, 4, 16\}$ , optionally up to 1024. An atom of order  $R_q$  has coefficient of variation  $R_q^{-1/2}$ , so high orders give sharp peaks in log-time. From  $\mathbf{h}_n$  the network outputs:

- weights  $\pi_1, \dots, \pi_Q, \pi_\infty \geq 0$  summing to one, where  $\pi_\infty$  is the weight of a *defect* atom meaning “the renewal channel produces no event”;

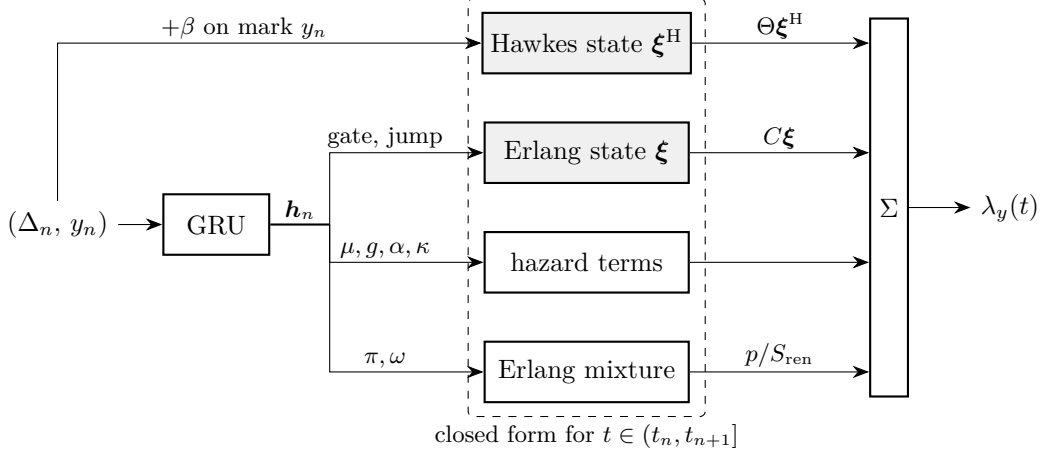


Figure 1: EPT-X at one event. The GRU summary  $\mathbf{h}_n$  sets the jump of the neural Erlang state, the hazard parameters and the renewal weights, while the Hawkes state is driven by the event mark alone. The grey boxes are the states carried from one event to the next. Until the next event, all four terms evolve in closed form, and their sum is the intensity  $\lambda_y(t)$ .

- for each time scale, a distribution  $\omega_{y|q}$  over the marks, so that the mark of a burst may depend on how quickly it follows the last event;
- optionally, a bounded shift of each log-rate, which lets an atom move within its grid cell.

The renewal channel's joint density of the next gap  $s$  and mark  $y$ , and its survival function, are

$$p(s, y) = \sum_{q=1}^Q \pi_q \omega_{y|q} \frac{\beta_q^{R_q} s^{R_q-1} e^{-\beta_q s}}{(R_q - 1)!}, \quad S_{\text{ren}}(s) = \pi_\infty + \sum_{q=1}^Q \pi_q e^{-\beta_q s} \sum_{i=0}^{R_q-1} \frac{(\beta_q s)^i}{i!}. \quad (7)$$

EPT-X adds the corresponding hazard to EPT:

$$\lambda_y(t) = \lambda_y^{\text{EPT}}(t) + \frac{p(s, y)}{S_{\text{ren}}(s)}.$$

Figure 1 shows how the parts fit together. The network only acts at event times; between events every term evolves in closed form.

**Proposition 5.** *Let  $\Lambda^{\text{EPT}}(\Delta) = \int_{t_n}^{t_n+\Delta} \sum_y \lambda_y^{\text{EPT}}(t) dt$  be the compensator of EPT over a gap of length  $\Delta$ .*

1. *The compensator of EPT-X over the same gap is  $\Lambda^{\text{EPT}}(\Delta) - \log S_{\text{ren}}(\Delta)$ , and both terms are in closed form.*
2. *If  $\pi_\infty = 1$ , EPT-X coincides with EPT, which in turn contains the linear phase-type Hawkes process.*
3. *Given  $\mathbf{h}_n$ , the gap produced by the renewal channel is the absorption time of a finite continuous-time Markov chain, so its survival function, quantiles and moments are closed-form matrix expressions.*
4. *As the grid of atoms is refined, the renewal channel can approximate any conditional distribution of the next gap.*

*Proof. Part 1.* The intensity of EPT-X is a sum of two intensities, so its compensator is the sum of the two compensators. For the renewal channel,  $\sum_y p(s, y) = -\frac{d}{ds} S_{\text{ren}}(s)$ , because each Erlang density is the negative derivative of its survival function. Hence

$$\int_0^\Delta \frac{\sum_y p(s, y)}{S_{\text{ren}}(s)} ds = - \int_0^\Delta \frac{d}{ds} \log S_{\text{ren}}(s) ds = -\log S_{\text{ren}}(\Delta),$$

using  $S_{\text{ren}}(0) = 1$ . The EPT part is closed form by Lemma 1 and by direct integration of the two exponential hazard terms.

*Part 2.* If  $\pi_\infty = 1$ , all other weights are zero. Then  $p \equiv 0$  and  $S_{\text{ren}} \equiv 1$ , so the renewal hazard vanishes. Setting the neural read-outs of EPT to zero leaves the Hawkes term.

*Part 3.* An Erlang distribution of order  $R$  is the absorption time of a chain of  $R$  exponential phases. A mixture of such chains, entered at random according to the weights  $\pi_q$ , together with an absorbing “never” state of weight  $\pi_\infty$ , is again a finite continuous-time Markov chain with one absorbing state.

*Part 4.* Mixtures of Erlang distributions are dense in the distributions on  $(0, \infty)$  [11, 17]. The renewal channel is such a mixture, with weights that depend on the history. This is the same expressiveness as a conditional mixture density [15], with the difference that the intensity and compensator stay exact.  $\square$

**Training and prediction.** We maximise the exact log-likelihood  $\sum_n \log \lambda_{y_n}(t_n) - \Lambda$ , where  $\Lambda$  is the total compensator, so the gradients contain no sampling noise. For large orders we evaluate  $S_{\text{ren}}$  through the regularised upper incomplete gamma function, and we take care that its gradient stays finite at zero gaps. The log-likelihood splits exactly into a time term,  $\log \lambda(t_n) - \Lambda$  with  $\lambda = \sum_y \lambda_y$ , and a mark term,  $\log(\lambda_{y_n}(t_n)/\lambda(t_n))$ , which we report separately. Predictions of the next event are Bayes-optimal functionals of the exact density: the expected gap is  $\int_0^\infty S(s) ds$ , where  $S$  is the survival function of the full model, and the predicted mark is  $\arg \max_y \int_0^\infty \lambda_y(s) S(s) ds$ . Both integrals are computed by one-dimensional quadrature on a logarithmic grid.

## 5 Data

Figure 2 summarises the steps from raw data to absorption times.

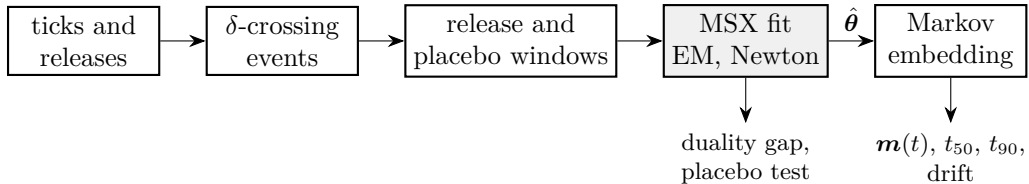


Figure 2: Measurement pipeline. Prices become  $\delta$ -crossing events, which are grouped into release windows and matched placebo windows. A single certified MSX fit gives the duality gap and the placebo test, and its Markov embedding gives the response  $\mathbf{m}(t)$ , the absorption times and the expected price drift in closed form.

**Releases.** The calendar covers January 2022–August 2026 (670 scheduled events). US data releases (CPI, nonfarm payrolls, PPI, retail sales, GDP advance estimate, initial jobless claims) are dated by their *initial* vintage in ALFRED, which removes seasonal-factor-only releases automatically; all are released at 08:30 America/New\_York and converted to UTC with daylight saving.

FOMC statements (14:00 ET) and press conferences (14:30 ET) come from the Federal Reserve calendar; ECB decisions and press conferences from the ECB (13:45/14:30 CET until 21 July 2022 and 14:15/14:45 CET afterwards); Bank of Japan decisions from the BoJ, whose release time is not fixed (a latent onset, flagged).

**Expectations and surprises.** CPI expectations are the Cleveland Fed inflation nowcast for the reference month, taken from the last nowcast strictly before the release date; GDP expectations are the Atlanta Fed GDPNow forecast immediately before BEA’s advance estimate. For payrolls, claims, PPI and retail sales we use real-time statistical expectations from the sequence of first prints (mean of the previous 3, 1, 6 and 6 prints respectively); survey consensus is not freely available and this limitation is stated wherever surprises are used. Surprises are standardised by the expanding standard deviation of past surprises of the same type (no look-ahead) and winsorised at  $\pm 4$ .

**Prices.** FX (EUR/USD, USD/JPY), gold (XAU/USD) and an S&P 500 CFD come from HistData millisecond bid/ask ticks (stamped in fixed EST, converted to UTC); BTC and ETH from Binance USD<sup>M</sup>-futures aggregated trades. A clock check at a CPI release confirms activity jumps within one second of the scheduled time in both sources. The HistData feed has a gap from February to July 2023 for all four instruments; windows without at least 90% of 10-minute blocks containing five ticks are excluded (per window, documented in the data card).

**Events and windows.** Prices are converted to  $\delta$ -crossing events with  $\delta$  calibrated per asset so that the median intraday rate is about 0.2 events per second:  $\delta = 0.25$  (EUR/USD, USD/JPY), 0.7 (XAU/USD, S&P 500 CFD), 2.0 (BTC) and 3.0 (ETH) basis points; simultaneous crossings collapse into one event with a size mark. Releases closer than 60 minutes are merged into one window [first  $-30$  min, last  $+60$  min] with 30 minutes of burn-in history. Each window is matched with a placebo window at the same clock times on the same weekday one to six weeks away, with no modelled release and none of sixteen second-tier US releases within three hours; its pseudo-releases carry a per-type PLACEBO type.

## 6 Benchmark protocol

All result tables are regenerated from JSON files written by the scripts in `scripts/`; every method is run by us on identical data unless a table explicitly cites published numbers.

**Track A: synthetic recovery.** Five truths, simulated by exact cluster sampling, deliberately include kernels *outside* the MSX class: S1 exponential (the home turf of single-decay estimators), S2 a true power law (Lomax, tail exponent 1.5, as reported for FX and futures), S3 a gamma hump plus a fast exponential, S4 a sparse 4-dimensional multiscale network (10 ms and 30 s components), S5 a sparse 10-dimensional power-law network. Each scenario uses 10 windows per train and test set and five seeds. Competitors: the exponential-kernel MLE of `tick` [4] (decay chosen by training likelihood), sum-of-exponentials MLE on the same decay grid as MSX, `tick`’s nonparametric EM with histogram kernels, ADM4 [24], the conditional-law estimator [3] (quadrature chosen by training likelihood), and NPHC [1] (our PyTorch solver; `tick` 0.8 ships a stub). Metrics: relative Frobenius error of the branching matrix, spectral-radius error, relative  $L^1$  kernel error on lags  $10^{-4}$ – $10^4$  s, edge AUC/AP, and held-out log-likelihood deficit per event relative to the true model ( $dLL$ , lower is better). MSX-auto chooses the Erlang order and an exposure-weighted  $\ell_1$  level on held-out training windows only.

**Track B: public marked-TPP benchmarks.** EasyTPP splits of Amazon, Retweet, Taxi, Taobao and StackOverflow and the protocol of S2P2 [5]: log-likelihood per scored event with the first event of each sequence as conditioning, model selection on validation likelihood, five seeds. Published numbers are quoted from Chang et al. [5]; our re-run of S2P2 validates the harness. EPT-TPP reports the *exact* likelihood; neural baselines estimate the compensator by Monte Carlo.

**Track C: limit-order-book event streams.** LOBSTER samples (AAPL, AMZN, GOOG, INTC, MSFT, 21 June 2012), six event types (limit orders, cancellations and market orders on each side), a chronological 60/20/20 split, sequences of 128 events. Neural baselines (RMTPP, NHP, THP, SAHP, AttNHP, IntensityFree, S2P2) run with the S2P2 authors’ configurations; on our 8 GB laptop GPU the batch size is reduced from 256 to 64 to avoid the Windows GPU watchdog. The classical Hawkes family is scored under the same protocol.

**Track D: post-release forecasting.** Chronological split of the news panel (train 2022–2024, test 2025–2026). Targets are the number of  $\delta$ -crossings (activity, and realised variance  $\delta^2 N$ ) and the signed drift in the 1, 5 and 15 minutes after each release. MSX forecasts are the closed form of Section 3 with the window baseline re-estimated from the pre-release data; baselines are same-type climatology, an event-study regression on pre-release activity and  $|z|$ , and a HAR regression on lagged activity.

## 7 Results

### 7.1 Track A: synthetic recovery

Table 1 reports the held-out likelihood deficit (means and standard deviations over five seeds); branching-matrix, spectral-radius and kernel errors are in Table 9 (Appendix). When the truth is a single exponential (S1), the estimators that assume exactly this form (tick’s exponential MLE and ADM4) are best. Under a true power law (S2), the sum-of-exponential estimators of tick and MSX perform about equally (held-out deficit 0.2 against  $0.3 \times 10^{-3}$  nats per event; tick’s solver failed on one of five seeds). When the kernel has structure that no single family captures, namely a delayed peak (S3), components ten milliseconds and thirty seconds apart (S4) or a sparse ten-dimensional power-law network (S5), MSX-auto has the smallest held-out deficit, 7 to 10 times smaller than the best competitor’s. On the two network scenarios it also roughly halves the best competitor’s branching-matrix error and recovers the causal graph perfectly ( $\text{AUC} = 1$ , as do several competitors); on the hump the nonparametric EM estimates the branching matrix slightly better. The exponential-only ablation isolates the contribution of Erlang phases, and the certificate of Proposition 2 was at most  $3.3 \times 10^{-4}$  nats over all 100 MSX fits.

Table 1: Track A, synthetic recovery: held-out log-likelihood deficit to the true model ( $10^{-3}$  nats per event, lower is better; mean  $\pm$  s.d. over five seeds). Bold: best; underlined: second best; shaded: our estimators.

|                             | S1 exp                 | S2 power law           | S3 hump                | S4 multiscale          | S5 network             |
|-----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <i>Classical estimators</i> |                        |                        |                        |                        |                        |
| Exp. kernel MLE             | <b>0.05</b> $\pm 0.05$ | 26.05 $\pm 0.50$       | 101.47 $\pm 1.81$      | 711.28 $\pm 53.09$     | 426.80 $\pm 368.57$    |
| Sum-of-exp. MLE             | 0.92 $\pm 0.13$        | <b>0.18</b> $\pm 0.12$ | 11.51 $\pm 0.46$       | —                      | —                      |
| ADM4                        | <u>0.05</u> $\pm 0.05$ | 26.05 $\pm 0.50$       | 101.47 $\pm 1.81$      | 3.13 $\pm 0.25$        | 26.51 $\pm 0.29$       |
| Hawkes EM (nonparam.)       | 1.02 $\pm 0.07$        | 1.73 $\pm 0.21$        | 2.24 $\pm 0.12$        | 4.55 $\pm 0.29$        | 5.23 $\pm 0.25$        |
| Conditional law             | 2.84 $\pm 1.03$        | 4.91 $\pm 0.59$        | 10.66 $\pm 2.13$       | 114.94 $\pm 1.20$      | 78.64 $\pm 4.79$       |
| <i>Ours</i>                 |                        |                        |                        |                        |                        |
| MSX ( $R = 2$ )             | 0.23 $\pm 0.12$        | 0.32 $\pm 0.12$        | 5.90 $\pm 0.36$        | 0.54 $\pm 0.12$        | 1.19 $\pm 0.07$        |
| MSX ( $R = 4$ )             | 0.28 $\pm 0.15$        | 0.45 $\pm 0.10$        | <u>0.31</u> $\pm 0.15$ | 0.70 $\pm 0.11$        | 1.63 $\pm 0.09$        |
| MSX, exponential only       | 0.91 $\pm 0.13$        | <u>0.19</u> $\pm 0.11$ | 11.51 $\pm 0.46$       | <u>0.32</u> $\pm 0.08$ | <u>0.82</u> $\pm 0.08$ |
| MSX-auto                    | 0.24 $\pm 0.12$        | 0.33 $\pm 0.13$        | <b>0.30</b> $\pm 0.13$ | <b>0.32</b> $\pm 0.08$ | <b>0.74</b> $\pm 0.10$ |

## 7.2 Main study: how fast do markets absorb scheduled macro news?

The pooled 2022–2026 model has 732 windows (430 release and 302 placebo windows) and twelve event types. It scores 7.3 million  $\delta$ -crossing events, or 8.4 million including the burn-in history, and its certificate bounds the distance to the optimum by 0.0054 nats per dimension. The spectral radius of the branching matrix is  $\rho(G) = 0.87$ , so roughly seven in every eight price events are triggered by earlier price events rather than directly by news. Fitted year by year,  $\rho(G)$  declines from 2022 to 2026 (0.91, 0.89, 0.85, 0.86, 0.85), and the slowest echo mode relaxes in about 12 minutes.

**Falsification.** Figure 3 compares, for every release type, the direct news-kernel mass with that of a matched placebo at the same clock time on a day without any scheduled release (neither the 11 modelled types nor sixteen second-tier US releases). With the exposure-weighted  $\ell_1$  level selected on held-out windows, 5 of the 11 release types exceed their placebo by at least 50%: CPI (8.5 $\times$ ), payrolls (6.7 $\times$ ), PPI (4.0 $\times$ ), the FOMC statement (3.9 $\times$ ) and retail sales (1.9 $\times$ ). Jobless claims, ECB and BoJ decisions and the press conferences show no detectable *direct* effect beyond their placebos. Without the penalty this negative result would have been hidden: nonnegative news kernels then absorb as much mass on placebo days as on release days, because estimates cannot fall below zero. Our first specification, with long news kernels, failed the test outright (Appendix).

**Echo-corrected absorption.** For the releases that pass the placebo test, the direct news kernel is absorbed almost instantly (median  $t_{50}$  of 1.1 s), but the market’s own echoes stretch the response: half of the total extra activity arrives within 16–137 s and ninety percent within 5–11 minutes (Figure 4, Table 2). The textbook “news half-life”  $\ln 2/\beta$  therefore understates absorption time by one to two orders of magnitude.

**Price discovery.** Signed surprises enter the news kernel separately by sign, so the model yields expected drift curves (Figure 5). The signs are economically coherent: hotter-than-expected inflation and stronger payrolls depress crypto and lift the dollar, while cooler prints do the opposite.

**Goodness of fit.** Time-rescaled residuals have Kolmogorov–Smirnov statistics of at most 0.044 against  $\text{Exp}(1)$ , and the per-window excess-dispersion test rejects in 26–55% of windows, compared with 42% (power-law kernel) and 93% (double exponential) reported for EBS FX data by Rambaldi et al. [14].

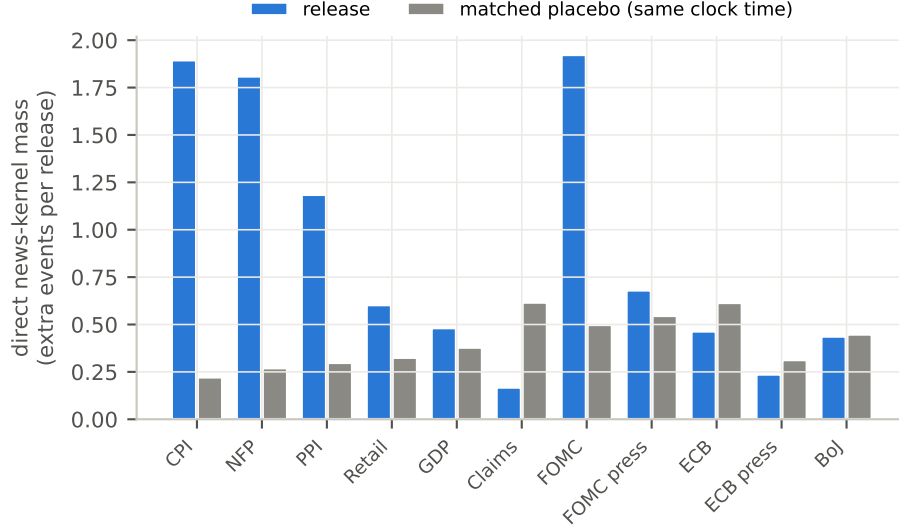


Figure 3: Placebo test. Direct news-kernel mass, the expected number of extra events per release summed over all assets, for each release type (blue) and for its matched placebo at the same clock time on a release-free day (grey). A type passes if its mass exceeds the placebo mass by at least 50%.

Table 2: Echo-corrected absorption times from the pooled model:  $t_{50}$  in seconds /  $t_{90}$  in minutes after the release. \*Release types whose news-kernel mass does not exceed their placebo by 50% (Figure 3).

| Release           | EURUSD     | USDJPY     | Gold     | S&P 500    | BTC      | ETH       |
|-------------------|------------|------------|----------|------------|----------|-----------|
| CPI               | 64 / 9.1   | 126 / 11.0 | 19 / 4.8 | 137 / 11.0 | 17 / 6.1 | 33 / 7.8  |
| NFP               | 94 / 10.1  | 109 / 10.7 | 16 / 5.2 | 80 / 8.8   | 27 / 7.5 | 24 / 7.3  |
| PPI               | 120 / 10.8 | 94 / 10.3  | 40 / 6.4 | 79 / 8.6   | 23 / 7.2 | 39 / 8.7  |
| FOMC              | 89 / 10.1  | 76 / 9.8   | 82 / 9.0 | 77 / 8.9   | 41 / 9.2 | 36 / 9.1  |
| Retail            | 59 / 9.1   | 77 / 9.8   | 87 / 9.2 | 114 / 10.4 | 26 / 7.6 | 26 / 7.8  |
| GDP*              | 95 / 10.0  | 81 / 9.6   | 26 / 5.1 | 145 / 11.2 | 27 / 7.3 | 27 / 7.5  |
| FOMC press conf.* | 88 / 10.0  | 111 / 10.9 | 60 / 8.1 | 59 / 8.0   | 45 / 9.5 | 52 / 10.0 |
| Claims*           | 124 / 11.0 | 56 / 9.1   | 87 / 9.3 | 81 / 9.1   | 36 / 9.0 | 40 / 9.4  |
| ECB*              | 61 / 9.2   | 73 / 9.6   | 88 / 9.0 | 123 / 10.8 | 37 / 9.2 | 42 / 9.6  |
| ECB press conf.*  | 137 / 11.3 | 90 / 10.2  | 38 / 6.5 | 63 / 8.2   | 25 / 7.3 | 32 / 8.0  |
| BoJ*              | 93 / 10.2  | 80 / 9.9   | 45 / 7.3 | 73 / 8.8   | 35 / 8.5 | 37 / 8.8  |

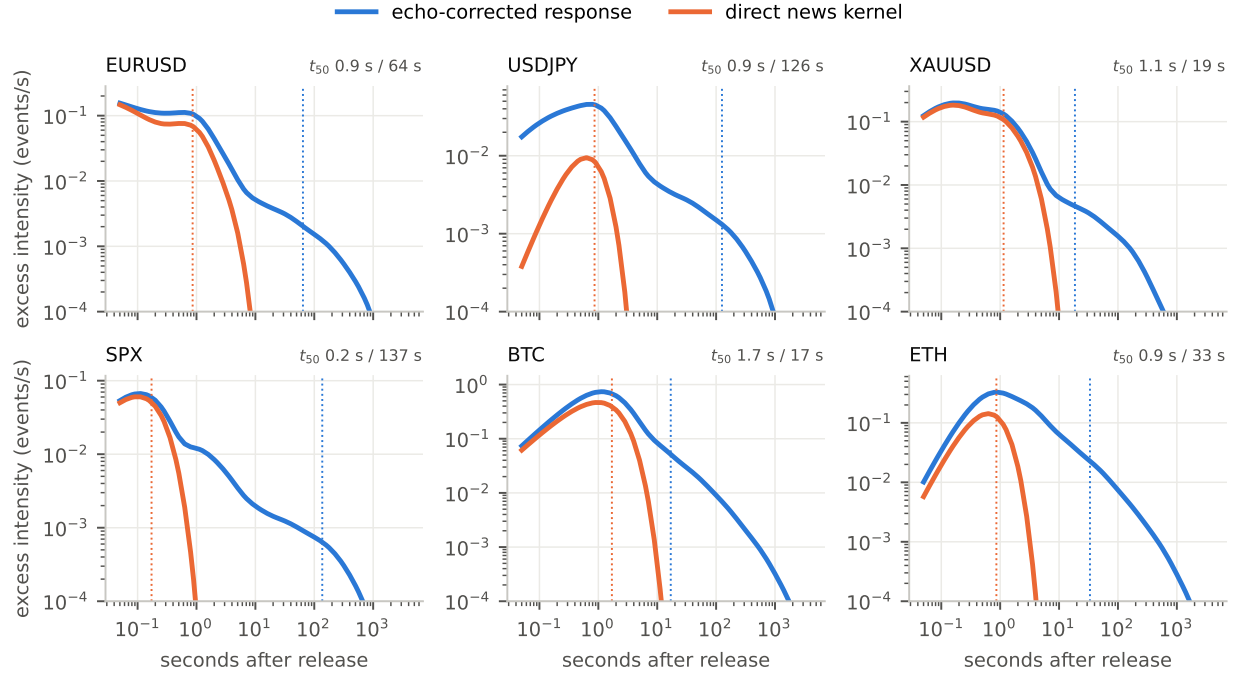


Figure 4: Response of each asset to a CPI release, in excess events per second (both axes logarithmic). Orange: direct news kernel. Blue: echo-corrected response  $\mathbf{m}(t)$ , which adds the events triggered by earlier events. Dotted lines and panel titles give  $t_{50}$ , direct / echo-corrected.

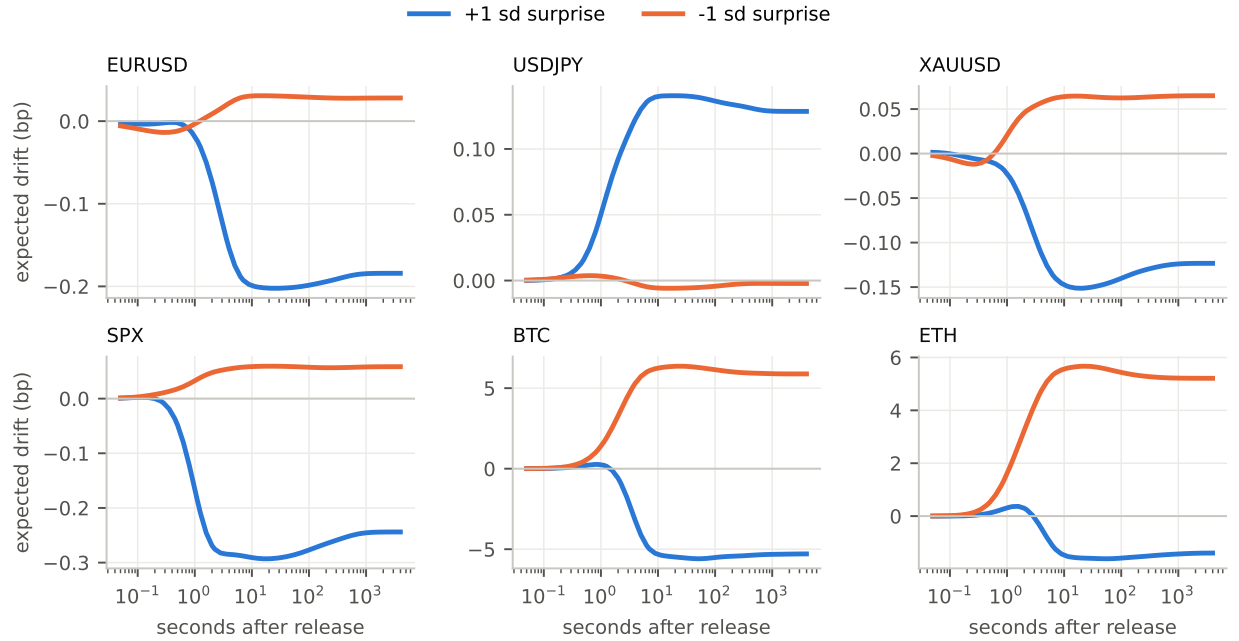


Figure 5: Expected cumulative price change, in basis points, after a CPI surprise of +1 (blue) or -1 (orange) standard deviation, from the pooled model. Time axis logarithmic.

Table 3: Expected price drift (basis points, end of window) after a +1 standard-deviation surprise, by release and asset (pooled model). \*As in Table 2.

| Release           | EURUSD | USDJPY | Gold | S&P 500 | BTC  | ETH  |
|-------------------|--------|--------|------|---------|------|------|
| CPI               | -0.2   | +0.1   | -0.1 | -0.2    | -5.3 | -1.4 |
| NFP               | -0.1   | +0.1   | +0.1 | +0.2    | -0.3 | -1.5 |
| PPI               | 0.0    | +0.1   | -0.2 | -0.3    | -2.4 | -2.1 |
| FOMC              | 0.0    | -0.1   | +0.1 | +0.1    | -0.3 | +0.2 |
| Retail            | +0.1   | -0.1   | +0.2 | 0.0     | -1.3 | -0.5 |
| GDP*              | +0.1   | -0.1   | -0.1 | 0.0     | +0.9 | +0.2 |
| FOMC press conf.* | 0.0    | 0.0    | 0.0  | +0.1    | 0.0  | +0.1 |
| Claims*           | 0.0    | 0.0    | 0.0  | 0.0     | +0.4 | +0.6 |
| ECB*              | 0.0    | 0.0    | 0.0  | 0.0     | -0.1 | -0.1 |
| ECB press conf.*  | 0.0    | 0.0    | 0.0  | 0.0     | -0.3 | -0.3 |
| BoJ*              | 0.0    | 0.0    | 0.0  | 0.0     | -0.1 | +0.3 |

### 7.3 Track B: public marked-TPP benchmarks

Table 4 compares our models with the published numbers of Chang et al. [5] under the same protocol; our re-run of S2P2 on Taxi (0.524 against the published 0.522) validates the harness. On Retweet, the largest dataset (2.2 million training events), EPT-X improves on the best published model by more than 0.6 nats per event, with a 25-epoch budget imposed by compute and the validation likelihood still rising. It is statistically tied with the best published model on Taobao and StackOverflow, and close on Taxi. There our time term equals S2P2’s (0.733 vs. 0.735 nats per event on the re-run) and the whole difference lies in the mark term. On Amazon, whose inter-event times concentrate in a narrow band (median 0.73, 99th percentile 0.80), the Erlang orders used elsewhere (coefficient of variation  $\geq 1/4$ ) cannot represent the mode; adding atoms of order up to 1024 (coefficient of variation  $1/32$ ), evaluated through the regularised incomplete gamma function, raises EPT-X by 1.1 nats per event on both validation and test, into second place behind S2P2. Neither a deeper (three residual layers) nor a wider (128 units) encoder improved the validation likelihood on any dataset. The renewal channel adds 0.07 nats per event over EPT on Taxi and 0.02 on StackOverflow. The linear MSX model is close to the neural models on Taobao (1.305) and Retweet, and far behind on Amazon and Taxi, where the dynamics are not self-exciting.

For next-event prediction (Table 5), EPT-X predicts the time of the next event as well as or better than every published model on all five datasets. Its accuracy for the mark of the next event is lower than that of S2P2 and NHP on four of them. This is the same split between time and mark that the likelihood decomposition shows.

Table 4: Track B, public benchmarks: test log-likelihood per event (higher is better). Published numbers are from Chang et al. [5] (five seeds each); our rows give mean  $\pm$  s.d. over seeds (EPT-X: three to five seeds; other rows one seed), with the EPT-X configuration selected per dataset on validation. Bold: best; underlined: second best; shaded: our models.

|                                                  | Amazon                   | Retweet                   | Taxi              | Taobao                   | StackOverflow             |
|--------------------------------------------------|--------------------------|---------------------------|-------------------|--------------------------|---------------------------|
| <i>Neural TPPs, published (Chang et al. [5])</i> |                          |                           |                   |                          |                           |
| RMTTPP                                           | −2.136                   | −7.098                    | 0.346             | 1.003                    | −2.480                    |
| SAHP                                             | −2.074                   | −6.708                    | 0.298             | 1.168                    | −2.341                    |
| THP                                              | −2.096                   | −6.659                    | 0.372             | 0.790                    | −2.338                    |
| IntensityFree                                    | 0.496                    | −10.344                   | 0.453             | <b>1.318</b>             | −2.233                    |
| MHP                                              | −2.091                   | −6.564                    | 0.370             | 0.636                    | −2.346                    |
| NHP                                              | 0.129                    | <u>−6.348</u>             | <u>0.514</u>      | 1.157                    | −2.241                    |
| AttNHP                                           | 0.484                    | −6.499                    | 0.493             | 1.259                    | −2.194                    |
| S2P2                                             | <b>0.781</b>             | −6.365                    | <b>0.522</b>      | 1.304                    | <b>−2.163</b>             |
| <i>Classical Hawkes, our runs</i>                |                          |                           |                   |                          |                           |
| ADM4                                             | −2.260                   | −6.832                    | −0.315            | 1.184                    | −2.473                    |
| Conditional law                                  | −                        | −8.317                    | −2.922            | −63.919                  | −16.140                   |
| Hawkes EM (nonparam.)                            | −1.002                   | −6.655                    | −0.308            | 1.269                    | −2.370                    |
| <i>Ours</i>                                      |                          |                           |                   |                          |                           |
| MSX ( $R = 2$ )                                  | −1.943                   | −6.569                    | −0.214            | 1.299                    | −2.292                    |
| MSX-auto                                         | −1.927                   | −6.568                    | −0.193            | 1.305                    | −2.292                    |
| EPT                                              | −1.203                   | −                         | 0.437             | 1.311                    | −2.186                    |
| EPT-X                                            | <u>0.640</u> $\pm 0.001$ | <b>−5.716</b> $\pm 0.024$ | 0.503 $\pm 0.004$ | <u>1.316</u> $\pm 0.005$ | <u>−2.166</u> $\pm 0.001$ |

Table 5: Next-event prediction on the public benchmarks: time RMSE (top, lower is better; Retweet in seconds) and mark accuracy (bottom, %, higher is better). Published numbers from Chang et al. [5], Table 2(b,c). Bold: best; underlined: second best; shaded: ours.

|                                     | Amazon            | Retweet               | Taxi                     | Taobao            | StackOverflow            |
|-------------------------------------|-------------------|-----------------------|--------------------------|-------------------|--------------------------|
| <i>Published (Chang et al. [5])</i> |                   |                       |                          |                   |                          |
| RMTTP                               | 0.338             | 16488                 | 0.283                    | 0.126             | 1.049                    |
| SAHP                                | 0.335             | 16102                 | 0.290                    | 0.126             | 1.031                    |
| THP                                 | 0.332             | 16268                 | 0.285                    | <b>0.125</b>      | 1.033                    |
| IntensityFree                       | <b>0.327</b>      | 16625                 | 0.362                    | <u>0.125</u>      | 1.340                    |
| MHP                                 | 0.329             | 16109                 | 0.284                    | 0.126             | 1.046                    |
| NHP                                 | 0.339             | <u>15911</u>          | 0.282                    | 0.126             | 1.019                    |
| AttNHP                              | 2.656             | 16171                 | 1.739                    | 0.130             | 1.256                    |
| S2P2                                | <u>0.327</u>      | 15987                 | <b>0.281</b>             | 0.126             | <u>1.014</u>             |
| <i>Ours</i>                         |                   |                       |                          |                   |                          |
| EPT-X                               | 0.327 $\pm$ 0.000 | <b>14336</b> $\pm$ 29 | <u>0.281</u> $\pm$ 0.001 | 0.125 $\pm$ 0.000 | <b>1.013</b> $\pm$ 0.002 |

|                                     | Amazon         | Retweet        | Taxi           | Taobao                | StackOverflow  |
|-------------------------------------|----------------|----------------|----------------|-----------------------|----------------|
| <i>Published (Chang et al. [5])</i> |                |                |                |                       |                |
| RMTTP                               | 30.8           | 53.4           | 91.4           | 60.9                  | 45.6           |
| SAHP                                | 32.4           | 57.5           | 91.4           | 60.5                  | 44.7           |
| THP                                 | 34.6           | 60.2           | 91.4           | 60.0                  | 46.6           |
| IntensityFree                       | 35.9           | 50.4           | 91.8           | 61.0                  | 45.6           |
| MHP                                 | 35.1           | 60.0           | 91.4           | 60.7                  | 46.5           |
| NHP                                 | <u>39.4</u>    | <b>61.4</b>    | <u>92.9</u>    | <b>61.5</b>           | 47.1           |
| AttNHP                              | 38.9           | 60.7           | 92.6           | 61.3                  | <b>48.2</b>    |
| S2P2                                | <b>40.7</b>    | <u>61.3</u>    | <b>93.1</b>    | 61.1                  | <u>47.5</u>    |
| <i>Ours</i>                         |                |                |                |                       |                |
| EPT-X                               | 35.9 $\pm$ 0.1 | 58.8 $\pm$ 1.1 | 91.7 $\pm$ 0.1 | <u>61.4</u> $\pm$ 0.1 | 45.5 $\pm$ 0.0 |

## 7.4 Track C: limit-order-book event streams

Table 6 reports test log-likelihood per event on the five LOBSTER stocks. EPT-X is the best model on every stock, ahead of MSX-auto, EPT and all seven neural baselines; among the stocks on which every baseline has finished, our models win 5 of 5, by 0.23–0.45 nats per event over the best neural baseline. One baseline needs a correction. EasyTPP’s IntensityFree replaces every gap below  $10^{-5}$  s by  $10^{-5}$  s before evaluating its log-normal mixture, which can then put an arbitrarily narrow component on that value. The result is not the likelihood of the observed gaps. On INTC and MSFT, 29% and 32% of the test gaps are this short (10 to 14% on the other stocks), and there the clamped model reports the highest numbers (row marked \*). Retrained and scored on the actual gaps (clamp  $10^{-12}$  s, same 1500-epoch budget), it loses 1.1 and 0.7 nats per event and falls below EPT-X, on INTC also below MSX. For scale, a denser MSX dictionary (40 instead of 12 rates) or sharper EPT-X atoms change the INTC validation likelihood by at most about 0.05. The patch is in the replication package, and the clamped numbers are not ranked. Among the remaining neural baselines, recurrent and attention models with Monte-Carlo compensators (NHP, THP, AttNHP) fall well behind the exact-likelihood phase-type models. Two implementation notes matter for fairness. First, LOBSTER timestamps contain ties, which EasyTPP rejects; all models are trained and scored on the same files in which tied messages are displaced by  $U(0, 100\text{ ns})$ . Second, on our 8 GB GPU the neural baselines run with batch sizes 32–64 instead of 256. Every neural model is trained with a 100-epoch budget and the strongest competitors additionally with longer budgets; for each model the table reports the budget with the best *validation* likelihood, so longer training can only help the baselines. The Hawkes backbone is essential for EPT: without it EPT underperforms plain MSX (superseded run, Appendix). The renewal channel of EPT-X adds 0.10–0.19 nats per event over EPT on every stock.

Table 6: LOBSTER order books: test log-likelihood per event (higher is better; mean  $\pm$  s.d. over seeds where several were run). Bold: best; underlined: second best; shaded: our models. \*As shipped, IntensityFree clamps gaps at  $10^{-5}$  s before scoring, so its score is not the likelihood of the observed data; it is greyed and not ranked. The IntensityFree row is retrained and scored on the actual gaps.

|                                        | AAPL                     | AMZN                     | GOOG                     | INTC                     | MSFT                     |
|----------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| <i>Classical Hawkes</i>                |                          |                          |                          |                          |                          |
| Exp. kernel MLE                        | 1.961                    | –                        | −0.311                   | –                        | –                        |
| Hawkes EM (nonparam.)                  | <b>3.370</b>             | 2.243                    | 2.851                    | 5.771                    | 6.258                    |
| ADM4                                   | 1.961                    | 1.067                    | 1.306                    | 4.787                    | 5.211                    |
| Conditional law                        | −1.018                   | −17.814                  | −32.784                  | 0.379                    | −1.270                   |
| <i>Neural TPPs (EasyTPP, our runs)</i> |                          |                          |                          |                          |                          |
| RMTPP                                  | 0.243                    | −0.686                   | −0.948                   | 2.882                    | 3.285                    |
| THP                                    | −0.081                   | −0.948                   | −1.075                   | 1.714                    | 1.855                    |
| SAHP                                   | 1.777                    | 0.462                    | −0.428                   | 3.915                    | 4.557                    |
| NHP                                    | 1.374                    | 0.842                    | 0.846                    | 3.874                    | 4.183                    |
| AttNHP                                 | 1.220                    | −0.168                   | 0.004                    | 3.497                    | 3.771                    |
| S2P2                                   | 2.315                    | 0.717                    | 0.990                    | 4.949                    | 5.416                    |
| IntensityFree                          | 3.108                    | 2.103                    | 2.726                    | 5.713                    | 6.377                    |
| IntensityFree* (as shipped)            | 3.260                    | 1.921                    | 2.679                    | 6.790                    | 7.070                    |
| <i>Ours</i>                            |                          |                          |                          |                          |                          |
| MSX ( $R = 2$ )                        | 3.326                    | 2.249                    | 2.870                    | 5.812                    | 6.307                    |
| MSX-auto                               | 3.361                    | <u>2.281</u>             | <u>2.883</u>             | 5.814                    | 6.310                    |
| EPT                                    | 3.380 $\pm$ 0.008        | 2.229 $\pm$ 0.003        | 2.806 $\pm$ 0.005        | 6.024 $\pm$ 0.007        | 6.530 $\pm$ 0.005        |
| EPT-X                                  | <b>3.534</b> $\pm$ 0.016 | <b>2.332</b> $\pm$ 0.005 | <b>2.992</b> $\pm$ 0.012 | <b>6.163</b> $\pm$ 0.002 | <b>6.666</b> $\pm$ 0.001 |

## 7.5 Track D: forecasting the post-release burst

On the 163 releases of 2025–2026, a model fitted on 2022–2024 forecasts the number of  $\delta$ -crossings in the first 1, 5 and 15 minutes (Table 7). The closed-form MSX forecast improves on same-type climatology at every horizon but is beaten by the event-study OLS and HAR regressions, which are fitted directly to this target; we report this negative result as is. For the *direction* of the post-release move the structural model is informative: the correlation between the forecast and the realised signed drift over the first minute is 0.17, against 0.06 for the best baseline (mean drift by release type and surprise sign).

Table 7: Post-release activity forecasts on the 163 test releases of 2025–2026, from models fitted on 2022–2024.  $N$  is the number of  $\delta$ -crossings after the release; QLIKE scores the forecast of realised variance. Mean over assets; lower is better. Bold: best; underlined: second best; shaded: our model.

| Method          | RMSE of $\log(1 + N)$ |              |              | QLIKE of realised variance |              |              |
|-----------------|-----------------------|--------------|--------------|----------------------------|--------------|--------------|
|                 | 1 min                 | 5 min        | 15 min       | 1 min                      | 5 min        | 15 min       |
| Climatology     | 1.440                 | 1.308        | 1.199        | <b>1.263</b>               | 0.916        | 0.701        |
| Event-study OLS | <b>1.140</b>          | <b>0.925</b> | <b>0.776</b> | 2.552                      | <b>0.580</b> | <b>0.372</b> |
| HAR             | <u>1.217</u>          | <u>0.985</u> | <u>0.820</u> | <u>1.467</u>               | <u>0.762</u> | <u>0.468</u> |
| MSX closed form | 1.322                 | 1.049        | 0.955        | 2.913                      | 1.036        | 0.518        |

## 8 Conclusion

Making every kernel a nonnegative combination of phase-type densities turns Hawkes estimation into a certified convex problem and turns news responses into matrix exponentials. Three findings stand out. First, the “half-life” of news is not a property of the news kernel: echoes lengthen absorption by one to two orders of magnitude, and the correction is available in closed form. Second, falsification matters: our first two specifications produced plausible-looking news effects that the placebo test exposed as artefacts of long kernels, ties and boundary bias. Third, a linear Hawkes backbone and a hyper-Erlang renewal channel make a neural point process exact and the most accurate model on all five order-book streams; benchmark comparisons must, however, check that every model scores the same observed gaps. Limitations: expectations for payrolls, claims, PPI and retail sales are statistical rather than survey-based; the FX feed has a gap in the first half of 2023; neural baselines were trained with reduced batch sizes on an 8 GB GPU; and on the public EasyTPP benchmarks EPT does not reach the state of the art (Section 7).

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## A Specification history and the placebo test

We report the specifications that *failed* the built-in falsification test, because the failures are informative and because the final specification was chosen by them, not by the results we hoped for. All superseded outputs are kept in `results/json/superseded/`.

**Version 1.** This version used news kernels with time scales up to 30 minutes and Erlang order 3, so that mean lags reached 90 minutes. It had a constant baseline in each window and time-of-day effects in UTC, and it allowed tied events: a trade that crossed several  $\delta$  levels produced several events with the same millisecond timestamp. The single placebo type absorbed 169 extra events per pseudo-release, which is comparable to real releases, and the residuals of the crypto series had Kolmogorov–Smirnov statistics up to 0.58. Two causes explain this. News kernels longer than the post-release window absorbed slow trends within the window, and ties, which made up 55% of bitcoin events, violate the assumption that events occur one at a time.

**Version 2.** This version kept one event per timestamp, with the number of crossed levels as a mark. It restricted news kernels to time scales between 0.1 and 300s, used a linear baseline within each window and modelled seasonality in New York time. The placebo mass fell to 18.6 and the residual fit improved by an order of magnitude, but the placebo effect was still significant under Wald standard errors. Splitting the placebo by time of day showed placebo mass at every release time, including 14:00 ET on days without an FOMC meeting. Unmodelled 08:30 releases therefore could not be the whole explanation.

**Version 3 (final).** This version made two further changes. First, placebo days must also avoid sixteen second-tier US releases, such as PCE, durable goods, trade, housing, JOLTS and consumer sentiment. Second, the news kernels carry an  $\ell_1$  penalty, weighted by exposure, whose level is chosen by the likelihood of held-out windows (Table 8). Without the penalty, each of the several hundred nonnegative weights has a positive expected value even when there is no effect, so release and placebo masses are inflated equally. The held-out likelihood peaks at  $\ell_1 = 0.8$ , which is also where release and placebo masses separate.

Table 8: Choice of the  $\ell_1$  penalty on the news kernels. Held-out log-likelihood on the 2024 windows relative to the best level (bold), and the mean news-kernel mass of release and placebo windows.

| $\ell_1$ level | held-out LL (rel.) | mean release mass | mean placebo mass |
|----------------|--------------------|-------------------|-------------------|
| 0.2            | −93.7              | 10.79             | 6.23              |
| 0.4            | −27.1              | 5.20              | 2.89              |
| 0.8            | <b>0.0</b>         | 1.81              | 1.03              |
| 1.6            | −20.5              | 0.44              | 0.37              |

## B EPT ablations

**Hawkes backbone.** Without the explicit linear Hawkes state, EPT reached 2.84 nats per event on LOBSTER AAPL (three seeds, 100 epochs), which is below the classical MSX estimator. The network had to rediscover mark-specific linear excitation through its gated jumps, and it did not manage to within the epoch budget. The backbone is a residual path that makes the model exactly MSX when the neural read-outs are zero. With it, EPT is better than every neural baseline on all five stocks and better than MSX on three of them (Table 6).

**Renewal channel and encoder.** Adding the mixture-of-Erlang renewal channel (EPT-X) improves EPT on every LOBSTER stock and on every public dataset where both were run. On validation, sharper Erlang atoms (orders up to 1024) help where the gap distribution has sharp peaks: by 1.1 nats per event on Amazon, whose gaps are concentrated in a narrow band, and by 0.05 on INTC. Replacing the GRU by a causal Transformer encoder [20] lowers the validation likelihood on all five datasets we tested, and neither a three-layer residual encoder nor a 128-unit encoder improves any dataset (Section 7).

## C Track A: all metrics

Table 9: Synthetic recovery, all metrics (mean  $\pm$  s.d. over five seeds): relative error of the branching matrix and of its spectral radius, kernel  $L^1$  error and held-out deficit dLL ( $10^{-3}$  nats/event), all lower is better, and edge AUC (higher is better). AUC needs absent true edges, so it is defined only for S4 and S5 (n/a for the complete graphs S1–S3). “–”: not produced by the method; NPHC estimates only the branching matrix. Missing rows are fits that did not converge. Bold: best; underlined: second best within each scenario; shaded: ours.

|                                      | $G$ error                | $\rho$ error             | kernel $L^1$             | dLL ( $10^{-3}$ )      | edge AUC                 |
|--------------------------------------|--------------------------|--------------------------|--------------------------|------------------------|--------------------------|
| <i>S1: exponential</i>               |                          |                          |                          |                        |                          |
| Exp. kernel MLE                      | <b>0.018</b> $\pm$ 0.007 | <b>0.005</b> $\pm$ 0.003 | <b>0.018</b> $\pm$ 0.008 | <b>0.05</b> $\pm$ 0.05 | n/a                      |
| Sum-of-exp. MLE                      | 0.065 $\pm$ 0.022        | 0.030 $\pm$ 0.013        | 0.165 $\pm$ 0.023        | 0.92 $\pm$ 0.13        | n/a                      |
| ADM4                                 | <u>0.018</u> $\pm$ 0.007 | <u>0.005</u> $\pm$ 0.003 | <u>0.018</u> $\pm$ 0.008 | <u>0.05</u> $\pm$ 0.05 | n/a                      |
| Hawkes EM (nonparam.)                | 0.035 $\pm$ 0.015        | 0.017 $\pm$ 0.007        | 0.173 $\pm$ 0.010        | 1.02 $\pm$ 0.07        | n/a                      |
| NPHC (cumulants)                     | 0.247 $\pm$ 0.087        | 0.032 $\pm$ 0.024        | –                        | –                      | n/a                      |
| Conditional law                      | 0.120 $\pm$ 0.054        | 0.017 $\pm$ 0.016        | 0.193 $\pm$ 0.022        | 2.84 $\pm$ 1.03        | n/a                      |
| MSX-auto                             | 0.054 $\pm$ 0.032        | 0.020 $\pm$ 0.018        | 0.093 $\pm$ 0.035        | 0.24 $\pm$ 0.12        | n/a                      |
| MSX ( $R = 2$ )                      | 0.056 $\pm$ 0.032        | 0.019 $\pm$ 0.018        | 0.092 $\pm$ 0.035        | 0.23 $\pm$ 0.12        | n/a                      |
| MSX ( $R = 4$ )                      | 0.056 $\pm$ 0.035        | 0.023 $\pm$ 0.018        | 0.105 $\pm$ 0.034        | 0.28 $\pm$ 0.15        | n/a                      |
| MSX, exponential only                | 0.055 $\pm$ 0.025        | 0.026 $\pm$ 0.015        | 0.157 $\pm$ 0.022        | 0.91 $\pm$ 0.13        | n/a                      |
| <i>S2: power law</i>                 |                          |                          |                          |                        |                          |
| Exp. kernel MLE                      | 0.299 $\pm$ 0.011        | 0.219 $\pm$ 0.007        | 0.465 $\pm$ 0.003        | 26.05 $\pm$ 0.50       | n/a                      |
| Sum-of-exp. MLE                      | <u>0.057</u> $\pm$ 0.012 | 0.016 $\pm$ 0.007        | <b>0.094</b> $\pm$ 0.011 | <b>0.18</b> $\pm$ 0.12 | n/a                      |
| ADM4                                 | 0.299 $\pm$ 0.011        | 0.219 $\pm$ 0.007        | 0.465 $\pm$ 0.003        | 26.05 $\pm$ 0.50       | n/a                      |
| Hawkes EM (nonparam.)                | 0.057 $\pm$ 0.014        | 0.037 $\pm$ 0.008        | 0.188 $\pm$ 0.006        | 1.73 $\pm$ 0.21        | n/a                      |
| NPHC (cumulants)                     | 0.153 $\pm$ 0.154        | 0.041 $\pm$ 0.047        | –                        | –                      | n/a                      |
| Conditional law                      | 0.102 $\pm$ 0.067        | 0.029 $\pm$ 0.019        | 0.291 $\pm$ 0.028        | 4.91 $\pm$ 0.59        | n/a                      |
| MSX-auto                             | 0.064 $\pm$ 0.029        | 0.037 $\pm$ 0.029        | 0.108 $\pm$ 0.022        | 0.33 $\pm$ 0.13        | n/a                      |
| MSX ( $R = 2$ )                      | 0.057 $\pm$ 0.021        | <u>0.015</u> $\pm$ 0.010 | 0.114 $\pm$ 0.015        | 0.32 $\pm$ 0.12        | n/a                      |
| MSX ( $R = 4$ )                      | 0.062 $\pm$ 0.014        | 0.016 $\pm$ 0.006        | 0.131 $\pm$ 0.017        | 0.45 $\pm$ 0.10        | n/a                      |
| MSX, exponential only                | <b>0.054</b> $\pm$ 0.022 | <b>0.014</b> $\pm$ 0.010 | <u>0.097</u> $\pm$ 0.011 | <u>0.19</u> $\pm$ 0.11 | n/a                      |
| <i>S3: hump</i>                      |                          |                          |                          |                        |                          |
| Exp. kernel MLE                      | 0.166 $\pm$ 0.008        | 0.053 $\pm$ 0.004        | 0.801 $\pm$ 0.004        | 101.47 $\pm$ 1.81      | n/a                      |
| Sum-of-exp. MLE                      | 0.038 $\pm$ 0.011        | 0.018 $\pm$ 0.005        | 0.419 $\pm$ 0.005        | 11.51 $\pm$ 0.46       | n/a                      |
| ADM4                                 | 0.166 $\pm$ 0.008        | 0.053 $\pm$ 0.004        | 0.801 $\pm$ 0.004        | 101.47 $\pm$ 1.81      | n/a                      |
| Hawkes EM (nonparam.)                | <b>0.024</b> $\pm$ 0.011 | <b>0.010</b> $\pm$ 0.004 | 0.187 $\pm$ 0.008        | 2.24 $\pm$ 0.12        | n/a                      |
| NPHC (cumulants)                     | 0.240 $\pm$ 0.096        | 0.045 $\pm$ 0.048        | –                        | –                      | n/a                      |
| Conditional law                      | 0.149 $\pm$ 0.062        | 0.017 $\pm$ 0.010        | 0.319 $\pm$ 0.050        | 10.66 $\pm$ 2.13       | n/a                      |
| MSX-auto                             | <u>0.032</u> $\pm$ 0.016 | 0.016 $\pm$ 0.007        | <u>0.080</u> $\pm$ 0.013 | <b>0.30</b> $\pm$ 0.13 | n/a                      |
| MSX ( $R = 2$ )                      | 0.069 $\pm$ 0.009        | 0.037 $\pm$ 0.004        | 0.347 $\pm$ 0.007        | 5.90 $\pm$ 0.36        | n/a                      |
| MSX ( $R = 4$ )                      | 0.038 $\pm$ 0.010        | 0.017 $\pm$ 0.005        | <b>0.074</b> $\pm$ 0.010 | <u>0.31</u> $\pm$ 0.15 | n/a                      |
| MSX, exponential only                | 0.034 $\pm$ 0.009        | <u>0.015</u> $\pm$ 0.004 | 0.416 $\pm$ 0.008        | 11.51 $\pm$ 0.46       | n/a                      |
| <i>S4: multiscale network</i>        |                          |                          |                          |                        |                          |
| Exp. kernel MLE                      | 0.173 $\pm$ 0.041        | 0.095 $\pm$ 0.025        | 1.635 $\pm$ 0.040        | 711.28 $\pm$ 53.09     | 1.000 $\pm$ 0.000        |
| ADM4                                 | 0.399 $\pm$ 0.003        | 0.255 $\pm$ 0.001        | 0.410 $\pm$ 0.001        | 3.13 $\pm$ 0.25        | 1.000 $\pm$ 0.000        |
| Hawkes EM (nonparam.)                | 0.153 $\pm$ 0.012        | 0.045 $\pm$ 0.010        | 0.358 $\pm$ 0.012        | 4.55 $\pm$ 0.29        | 1.000 $\pm$ 0.000        |
| NPHC (cumulants)                     | 1.030 $\pm$ 0.304        | 0.211 $\pm$ 0.186        | –                        | –                      | 0.588 $\pm$ 0.155        |
| Conditional law                      | 0.464 $\pm$ 0.286        | 0.139 $\pm$ 0.050        | 1.059 $\pm$ 0.074        | 114.94 $\pm$ 1.20      | 0.831 $\pm$ 0.087        |
| MSX-auto                             | <b>0.084</b> $\pm$ 0.014 | <u>0.016</u> $\pm$ 0.006 | <b>0.128</b> $\pm$ 0.026 | <b>0.32</b> $\pm$ 0.08 | 1.000 $\pm$ 0.000        |
| MSX ( $R = 2$ )                      | 0.093 $\pm$ 0.015        | 0.021 $\pm$ 0.005        | 0.173 $\pm$ 0.024        | 0.54 $\pm$ 0.12        | 1.000 $\pm$ 0.000        |
| MSX ( $R = 4$ )                      | 0.101 $\pm$ 0.019        | 0.027 $\pm$ 0.007        | 0.207 $\pm$ 0.035        | 0.70 $\pm$ 0.11        | <u>1.000</u> $\pm$ 0.000 |
| MSX, exponential only                | <u>0.084</u> $\pm$ 0.014 | <b>0.016</b> $\pm$ 0.006 | <u>0.128</u> $\pm$ 0.026 | <u>0.32</u> $\pm$ 0.08 | <b>1.000</b> $\pm$ 0.000 |
| <i>S5: 10-dim. power-law network</i> |                          |                          |                          |                        |                          |
| Exp. kernel MLE                      | 0.337 $\pm$ 0.153        | 0.103 $\pm$ 0.032        | 1.302 $\pm$ 0.897        | 426.80 $\pm$ 368.57    | 0.977 $\pm$ 0.050        |
| ADM4                                 | 0.167 $\pm$ 0.003        | 0.096 $\pm$ 0.003        | 0.319 $\pm$ 0.002        | 26.51 $\pm$ 0.29       | 1.000 $\pm$ 0.000        |
| Hawkes EM (nonparam.)                | <u>0.085</u> $\pm$ 0.008 | 0.045 $\pm$ 0.008        | 0.280 $\pm$ 0.010        | 5.23 $\pm$ 0.25        | 1.000 $\pm$ 0.000        |
| NPHC (cumulants)                     | 1.348 $\pm$ 0.477        | 0.520 $\pm$ 0.372        | –                        | –                      | 0.672 $\pm$ 0.042        |
| Conditional law                      | 0.461 $\pm$ 0.047        | 0.037 $\pm$ 0.026        | 0.641 $\pm$ 0.025        | 78.64 $\pm$ 4.79       | 0.950 $\pm$ 0.033        |
| MSX-auto                             | <b>0.045</b> $\pm$ 0.004 | <b>0.015</b> $\pm$ 0.005 | <b>0.079</b> $\pm$ 0.005 | <b>0.74</b> $\pm$ 0.10 | 1.000 $\pm$ 0.000        |
| MSX ( $R = 2$ )                      | 0.111 $\pm$ 0.013        | 0.040 $\pm$ 0.013        | 0.180 $\pm$ 0.017        | 1.19 $\pm$ 0.07        | 1.000 $\pm$ 0.000        |
| MSX ( $R = 4$ )                      | 0.122 $\pm$ 0.008        | 0.051 $\pm$ 0.013        | 0.218 $\pm$ 0.015        | 1.63 $\pm$ 0.09        | <u>1.000</u> $\pm$ 0.000 |
| MSX, exponential only                | 0.099 $\pm$ 0.014        | <u>0.030</u> $\pm$ 0.014 | <u>0.145</u> $\pm$ 0.017 | <u>0.82</u> $\pm$ 0.08 | <b>1.000</b> $\pm$ 0.000 |